

FDTT 704: Correlation and Regression

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Introduction

- **Correlation** describes the strength of the linear association between two variables.
- It takes values between -1 (perfect negative) and + 1 (perfect positive)
- A value of 0 indicates no linear association
- We use ρ to indicate the population correlation coefficient and R or r to indicate the sample correlation coefficient.

Covariance

- We have previously discussed the variance as a measure of uncertainty of a random variable

$$Var(X) = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)^2$$

- In order to define correlation we first need to define covariance, which is a generalization of variance to two random variables

$$Cov(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$

Cont'd

- Covariance is not a measure of uncertainty but rather a measure of the degree to which X and Y tend to be large (or small) at the same time or the degree to which one tends to be large while the other is small.
- The magnitude of the covariance is not very informative since it is affected by the magnitude of both X and Y . However, the sign of the covariance tells us something useful about the relationship between X and Y .

Properties of Covariance

- $\text{Cov}(X,X) = \text{Var}(X)$
- $\text{Cov}(X,Y) = \text{Cov}(Y,X)$
- $\text{Cov}(X,Y) = 0$ if X and Y are independent
- $\text{Cov}(X,c) = 0$
- $\text{Cov}(aX,bY) = ab \text{Cov}(X,Y)$
- $\text{Cov}(X+a,Y+b) = \text{Cov}(X,Y)$
- $\text{Cov}(X,Y+Z) = \text{Cov}(X,Y) + \text{Cov}(X,Z)$

Correlation

- Since $Cov(X,Y)$ depends on the magnitude of X and Y we would prefer to have a measure of association that is not affected by changes in the scales of the variables.
- The most measure of linear association is correlation which is defined as:

$$\rho(X, Y) = \frac{Cov(X, Y)}{\sigma_X, \sigma_Y}$$

- Where the magnitude of the correlation measures the strength of the linear association and the sign determines if it is a positive or negative relationship.

Correlation and Independence

- Given random variables X and Y

X and Y are independent $\Rightarrow Cov(X, Y) = \rho(X, Y) = 0$

$$Cov(X, Y) = \rho(X, Y) = 0$$

does not imply X and Y are independent.

Example

A study is conducted involving 10 students to investigate the association between statistics and science tests. The question arises here; is there a relationship between the degrees gained by the 10 students in statistics and science tests?

Student	1	2	3	4	5	6	7	8	9	10
Statistics	20	23	8	29	14	12	11	20	17	18
Science	20	25	11	24	23	16	12	21	22	26

Solution

Statistics	Science					
x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
20	20	2.7	7.29	0	0	0
23	25	5.7	32.49	5	25	28
8	11	-9.3	86.49	-9	81	83
29	24	11.7	136.89	4	16	46
14	23	-3.3	10.89	3	9	-9.9
12	16	-5.3	28.09	-4	16	21.2
11	12	-6.3	39.69	-8	64	50.4
21	21	3.7	13.69	1	1	3.7
17	22	-0.3	0.09	2	4	-0.6
18	26	0.7	0.49	6	36	4.2
173	200	0	356.1	0	252	228

$$\sum (x - \bar{x})^2 = 356.1, \quad \sum (y - \bar{y})^2 = 252,$$

$$\sum (x - \bar{x})(y - \bar{y}) = 228$$

Calculating the Pearson correlation coefficient;

$$\begin{aligned}
 r &= \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}} = \frac{228}{\sqrt{356.1} \sqrt{252}} \\
 &= \frac{228}{(18.8706)(15.8745)} = \frac{228}{299.5614} = 0.761
 \end{aligned}$$

In R

```
1 x <- c(20,23,8,29,14,12,11,21,17,18)
2
3 y <- c(20,25,11,24,23,16,12,21,22,26)
4
5 cor(x,y)

[1] 0.761112
```

Regression Analysis

- Regression analysis is one of the most commonly used statistical techniques in social and behavioral sciences as well as in physical sciences which involves identifying and evaluating the relationship between a dependent variable and one or more independent variables, which are also called predictor or explanatory variables.
- It is particularly useful for assess and adjusting for confounding.
- Model of the relationship is hypothesized and estimates of the parameter values are used to develop an estimated regression equation.
- When there is a single continuous dependent variable and a single independent variable, the analysis is called a simple linear regression analysis.

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- Multiple regression is to learn more about the relationship between several independent or predictor variables and a dependent or criterion variable.
- Independent variables are characteristics that can be measured directly; these variables are also called predictor or explanatory variables used to predict or to explain the behavior of the dependent variable.
- Dependent variable is a characteristic whose value depends on the values of independent variables.

Assumptions of Regression Analysis

- The relationship between independent variable and dependent is linear.
- The expected value of the error term is zero
- The variance of the error term is constant for all the values of the independent variable, the assumption of homoscedasticity.

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- There is no autocorrelation.
- The independent variable is uncorrelated with the error term.
- The error term is normally distributed.

Simple Linear Regression Model

- Simple linear regression is a statistical method that allows us to summarize and study relationships between two continuous (quantitative) variables.
- Least squares linear regression is a method for predicting the value of a dependent variable y , based on the value of an independent variable x .

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- One variable denoted by X is called the **predictor, explanatory or independent variable**
- The other variable denoted by Y is regarded as the **response, outcome or dependent variable**.

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- Mathematically, the regression model is represented by the following equation

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- The coefficients are estimated as follows:

$$\hat{\beta}_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$\hat{\beta}_0 = \bar{y} - \beta_1 \bar{x}$$

Example

Example – linear Regression of patient's age and their blood pressure: A study is conducted involving 10 patients to investigate the relationship and effects of patient's age and their blood pressure.

Obs	Age	BP	xy	x^2	<i>Required calculation</i> $\sum x = 491$ $\sum y = 1410$ $\sum xy = 71566$ $\sum x^2 = 26157$
	x	y			
1	35	112	3920	1225	
2	40	128	5120	1600	
3	38	130	4940	1444	
4	44	138	6072	1936	
5	67	158	10586	4489	
6	64	162	10368	4096	
7	59	140	8260	3481	
8	69	175	12075	4761	
9	25	125	3125	625	
10	50	142	7100	2500	
Total	491	1410	71566	26157	

Cont'd

- The mean $\bar{x} = 49.1$ and $\bar{y} = 141$
- The estimates

$$\beta_1 = \frac{10 \times 71566 - 491 \times 1410}{10 \times 26157 - (491)^2} = 1.140$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 141 - 55.974 = 85.026$$

- The fitted model is:

$$\hat{Y} = 85.026 + 1.140X$$

Cont'd

- The constant value $\beta_0 = 85.026$ indicates that blood pressure at age zero
- The $\beta_1 = 1.140$ indicates that as age increase by one year the blood pressure increase by 1.140
- Applying the value of age to the regression Model to calculate the estimated blood pressure ($\hat{Y}$) coefficient of determination (R^2) as follows:

Cont'd

Obs	Age	BP							
	X	Y	$\hat{Y}$	$\hat{Y} - \bar{y}$	$(\hat{Y} - \bar{y})^2$	$(y - \hat{Y})$	$(y - \hat{Y})^2$	$(y - \bar{y})$	$(y - \bar{y})^2$
1	35	112	124.926	-16.074	258.373	-12.926	167.081	-29	841
2	40	128	130.626	-10.374	107.620	-2.626	6.896	-13	169
3	38	130	128.346	-12.654	160.124	1.654	2.736	-11	121
4	44	138	135.186	-5.814	33.803	2.814	7.919	-3	9
5	67	158	161.406	20.406	416.405	-3.406	11.601	17	289
6	64	162	157.986	16.986	288.524	4.014	16.112	21	441
7	59	140	152.286	11.286	127.374	-12.286	150.946	-1	1
8	69	175	163.686	22.686	514.655	11.314	128.007	34	1156
9	25	125	113.526	-27.474	754.821	11.474	131.653	-16	256
10	50	142	142.026	1.026	1.053	-0.026	0.001	1	1
Total	491	1410	1410	0.000	2662.750	0.000	622.950	0	3284

Cont'd

- Equation of ANOVA table for simple linear regression;

<i>Source of Variation</i>	<i>Sums of Squares</i>	<i>Df</i>	<i>Mean Square</i>	<i>F</i>
<i>Regression</i>	$\Sigma(\hat{Y} - \bar{Y})^2$	<i>1</i>	$SS_{reg} / 1$	MS_{reg} / MS_{res}
<i>Residual</i>	$\Sigma(Y - \hat{Y})^2$	$N - 2$	$SS_{res} / (N - 2)$	
<i>Total</i>	$\Sigma(Y - \bar{Y})^2$	$N - 1$		

Cont'd

<i>Source of Variation</i>	<i>Sum of Squares</i>	<i>Df</i>	<i>Mean Square</i>	<i>F</i>
<i>Regression</i>	<i>2662.75</i>	<i>1</i>	<i>2662.75 / 1 = 2662.75</i>	<i>2662.75 / 77.86875 = 34.195</i>
<i>Residual</i>	<i>622.95</i>	<i>8</i>	<i>622.95 / 8 = 77.86875</i>	
<i>Total</i>	<i>3284</i>	<i>9</i>		

Calculating the coefficient of determination (R^2)

$$R^2 = \frac{\text{Explained Variation}}{\text{Total Variation}} = \frac{\text{Regression Sum of Square (SSR)}}{\text{Total Sum of Square (SST)}}$$

Then substitute the values from ANOVA table

$$R^2 = \frac{2662.75}{3284} = 0.810$$

We can say that 81% of the variation in the blood pressure rate is explained by age.